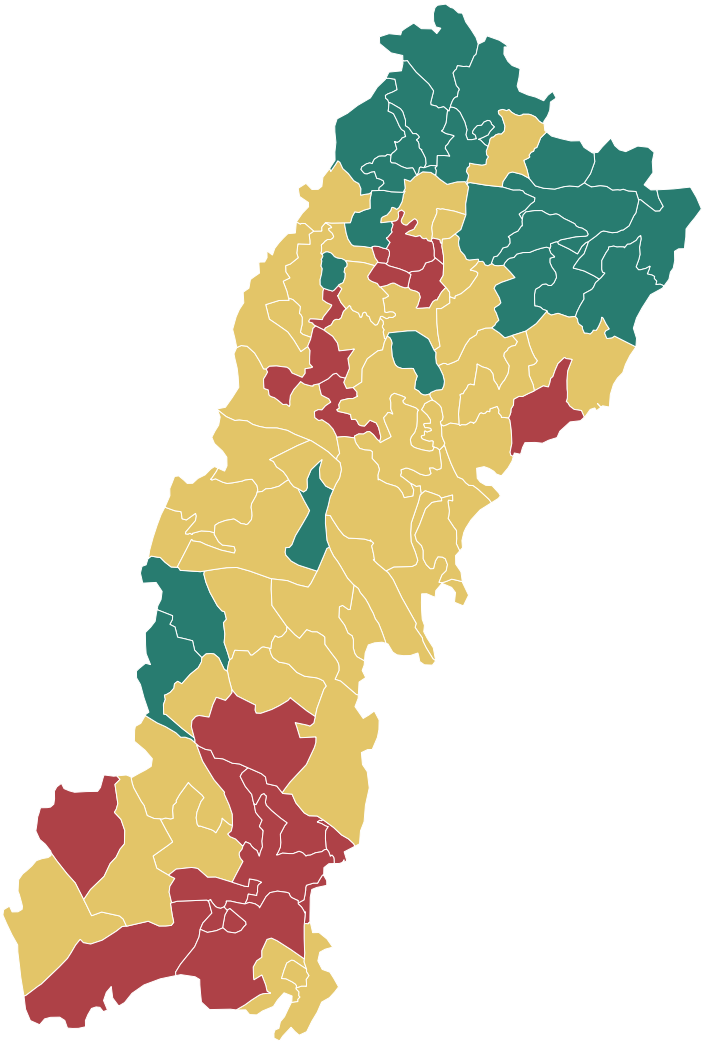
Spatiotemporal Urban Heat Vulnerability Assessment —

Multan, Pakistan (2003–2023)

2003 | Relative environmental screening



Grid N
↑

0 10 20 km

Within-year category

2003 — relative screening

- Lower
- Middle
- Higher

88 source polygons
82 council codes

Score components
LST percentile: 50%
NDBI percentile: 30%
Vegetation deficit: 20%

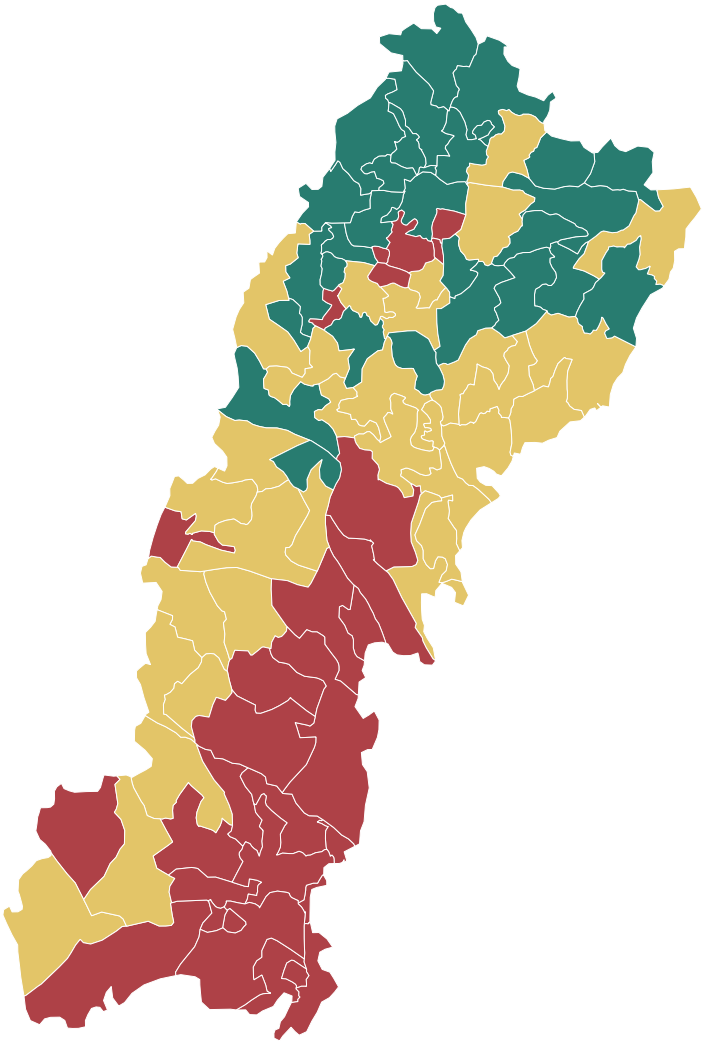
Lower: score ≤ 1/3
Middle: 1/3 < score ≤ 2/3
Higher: score > 2/3

Compare locations within this year. Colors are not health-risk thresholds or calibrated changes through time.
Sources: supplied Landsat / Earth Engine exports and Multan_UCs boundaries. Display CRS: WGS 84 / UTM 42N.

Spatiotemporal Urban Heat Vulnerability Assessment —

Multan, Pakistan (2003–2023)

2008 | Relative environmental screening



Grid N
↑

0 10 20 km

Within-year category

2008 — relative screening

- Lower
- Middle
- Higher

88 source polygons
82 council codes

Score components
LST percentile: 50%
NDBI percentile: 30%
Vegetation deficit: 20%

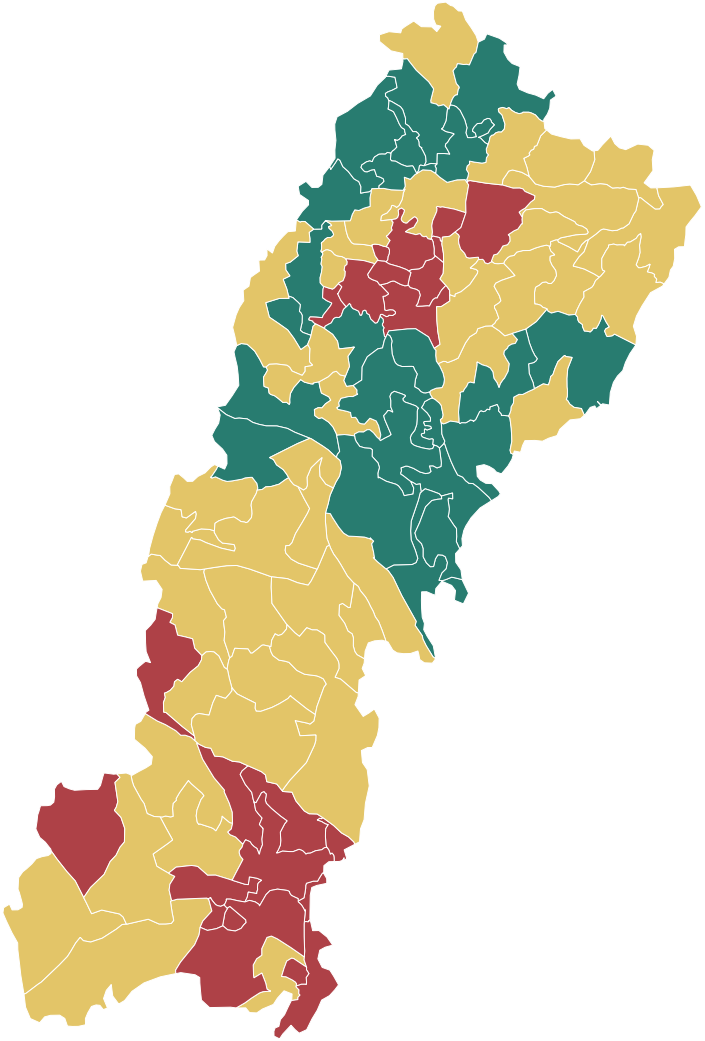
Lower: score ≤ 1/3
Middle: 1/3 < score ≤ 2/3
Higher: score > 2/3

Compare locations within this year. Colors are not health-risk thresholds or calibrated changes through time.
Sources: supplied Landsat / Earth Engine exports and Multan_UCs boundaries. Display CRS: WGS 84 / UTM 42N.

Spatiotemporal Urban Heat Vulnerability Assessment —

Multan, Pakistan (2003–2023)

2013 | Relative environmental screening



Grid N
↑

0 10 20 km

Within-year category

2013 — relative screening

- Lower
- Middle
- Higher

88 source polygons
82 council codes

Score components
LST percentile: 50%
NDBI percentile: 30%
Vegetation deficit: 20%

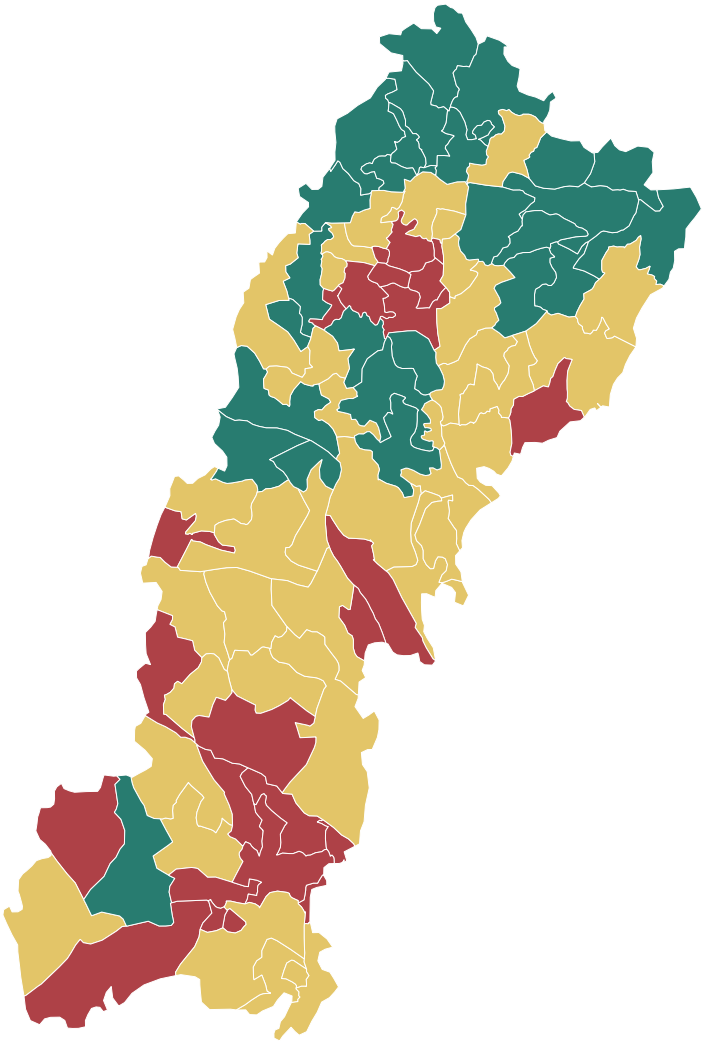
Lower: score ≤ 1/3
Middle: 1/3 < score ≤ 2/3
Higher: score > 2/3

Compare locations within this year. Colors are not health-risk thresholds or calibrated changes through time.
Sources: supplied Landsat / Earth Engine exports and Multan_UCs boundaries. Display CRS: WGS 84 / UTM 42N.

Spatiotemporal Urban Heat Vulnerability Assessment —

Multan, Pakistan (2003–2023)

2018 | Relative environmental screening



Grid N
↑

0 10 20 km

Within-year category

2018 — relative screening

- Lower
- Middle
- Higher

88 source polygons
82 council codes

Score components
LST percentile: 50%
NDBI percentile: 30%
Vegetation deficit: 20%

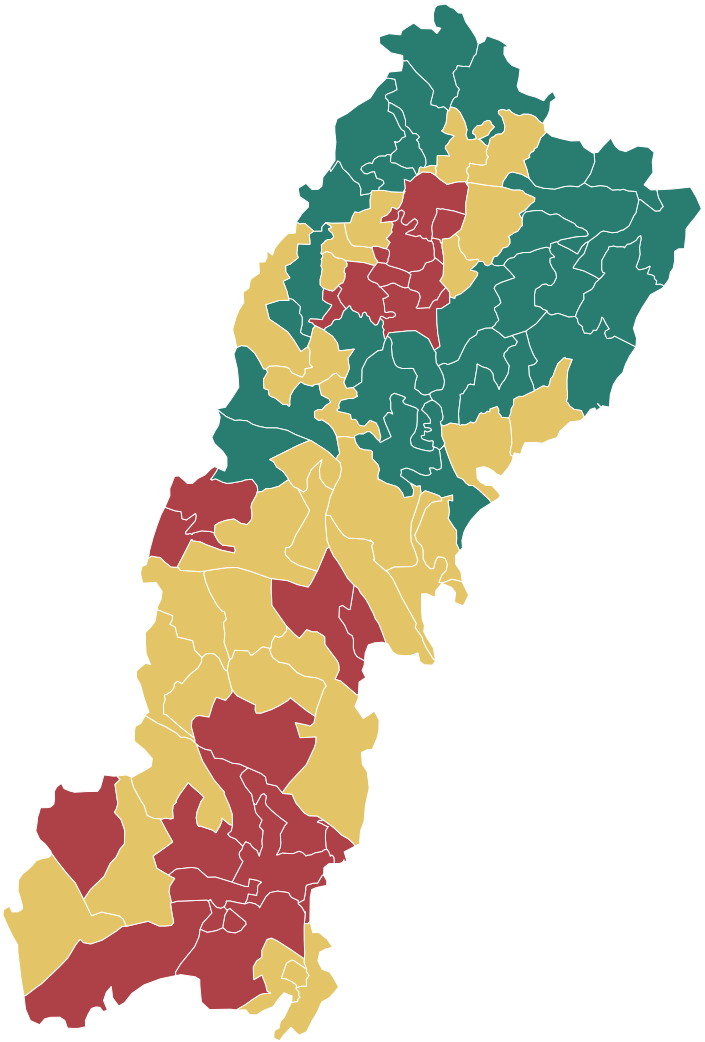
Lower: score ≤ 1/3
Middle: 1/3 < score ≤ 2/3
Higher: score > 2/3

Compare locations within this year. Colors are not health-risk thresholds or calibrated changes through time.
Sources: supplied Landsat / Earth Engine exports and Multan_UCs boundaries. Display CRS: WGS 84 / UTM 42N.

Spatiotemporal Urban Heat Vulnerability Assessment —

Multan, Pakistan (2003–2023)

2023 | Relative environmental screening



Grid N
↑

0 10 20 km

Within-year category

2023 — relative screening

- Lower
- Middle
- Higher

88 source polygons
82 council codes

Score components
LST percentile: 50%
NDBI percentile: 30%
Vegetation deficit: 20%

Lower: score ≤ 1/3
Middle: 1/3 < score ≤ 2/3
Higher: score > 2/3

Compare locations within this year. Colors are not health-risk thresholds or calibrated changes through time.
Sources: supplied Landsat / Earth Engine exports and Multan_UCs boundaries. Display CRS: WGS 84 / UTM 42N.